

A Finite Element Analysis Comparing the Biomechanical Stability of Fascia-Wrapped versus PRF-Embedded Diced Cartilage Grafts in Dorsal Augmentation Rhinoplasty

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Abstract

Rhinoplasty remains one of the most commonly performed plastic surgeries, yet reconstruction of the nasal dorsum continues to present challenges. Diced cartilage (DC) grafts have become a cornerstone of modern dorsal augmentation, providing precise contouring with low rates of warping. To stabilize these grafts, surgeons commonly utilize either autologous cartilage particles wrapped in fascia (DC-F) or cartilage particles embedded within a platelet-rich fibrin matrix (DC-PRF). Despite widespread clinical adoption, objective biomechanical comparisons between these techniques have been limited, and their relative *in vivo* stability and long-term performance under physiological mechanical stresses remain unclear. This study aimed to comparatively assess the biomechanical stability of DC-F and DC-PRF grafts using Finite Element Analysis. Three-dimensional nasal dorsum models were constructed in ANSYS, incorporating the two graft types. Simulations were conducted at multiple post-operative time periods to reflect healing and integration processes, with material properties adjusted based on literature-derived data for tissue healing and PRF matrix degradation over time. Various loading conditions were applied to simulate clinically relevant forces. Outcomes measured included graft internal von Mises stress distribution and deformation. The initial and long-term stability of DC-F and DC-PRF were compared. These results provide a quantitative framework to inform surgical decision-making, potentially improving aesthetic outcomes and reducing revision rates.

Keywords: finite element analysis, dorsal augmentation rhinoplasty, diced cartilage grafts

Introduction

Rhinoplasty is a surgical procedure performed to improve nasal aesthetics and/or function. From a cosmetic standpoint, rhinoplasty aims to modify the size, shape, and contour of the nose, while functionally it can enhance nasal airflow and breathing. Although it remains one of the most frequently performed plastic surgeries worldwide, reconstruction of the nasal dorsum continues to present technical challenges. The complex anatomy of the nose and substantial inter-individual variability contribute to postoperative complications such as asymmetry, surface irregularities, and open-roof deformities.¹

Over the past several decades, numerous surgical strategies and grafting techniques have been developed to address these challenges. Among available options, autologous graft materials are generally preferred over nonautologous alternatives due to their lower risks of rejection and infection, as well as their superior biocompatibility and adaptability. Common autologous graft materials include bone, fascia, and cartilage. Among these, diced cartilage (DC) grafts have become a cornerstone of modern dorsal augmentation because they allow precise contouring and exhibit a reduced incidence of complications such as graft warping. Currently, the two most widely used diced cartilage techniques are autologous cartilage particles wrapped in fascia (DC-F) and cartilage particles embedded within a platelet-rich fibrin matrix (DC-PRF).¹

Under general anesthesia, the DC-F technique is typically performed using an endonasal approach.² Following elevation of the mucoperichondrium and exposure of the defect, autologous fascia—harvested from the temporalis fascia or fascia lata—and cartilage from septal, auricular, or costal sources are obtained. The harvested cartilage is diced or crushed into small particles.³ A strip of fascia is folded to form a pouch that contains the cartilage particles, creating a sealed, pad-like graft. The pouch is closed using continuous sutures, resulting in a

flexible yet mechanically stable DC-F construct. The graft is sized according to the defect dimensions and placed using an underlay technique. Fixation is achieved by anchoring the fascia strip at two points on the nasal root's golden point and the nasal tip framework to ensure stable positioning.⁴ The fascia is subsequently adjusted to maintain moderate tension, ensuring close adhesion to the upper surface of the cavity for optimal integration. This technique leverages the biocompatibility and adaptability of autologous tissues while minimizing complications such as graft warping and contour irregularities.

In the DC-PRF technique, diced cartilage particles are incorporated into a platelet-rich fibrin (PRF) matrix derived from the patient's autologous blood, forming a cohesive and moldable composite that can be precisely shaped or injected.⁵ PRF is produced by centrifuging freshly drawn whole blood without anticoagulants, allowing spontaneous fibrin polymerization and clot formation.⁶ Blood volumes of approximately 10–100 mL are collected via venipuncture and immediately centrifuged to prevent premature coagulation, as delayed processing can compromise component separation. Following centrifugation, the fibrin-rich clot is separated from the red blood cell layer and subsequently cut or molded to meet surgical requirements. PRF is an autologous, biocompatible, and cost-effective material that is easy to manipulate and enhances regenerative potential when combined with cartilage particles in DC-PRF grafting.

In summary, DC-F and DC-PRF demonstrate distinct material properties as grafting materials for surgical applications: DC-F forms a relatively solid construct, whereas DC-PRF results in a gelatinous, adhesive composite. These differences in mechanical properties influence graft–tissue interactions during the healing process and are expected to result in divergent healing responses and postoperative outcomes. At the final stage, the mechanical properties and graft–tissue interactions are expected to become more similar as the graft integrates with the surrounding tissue. Despite their widespread clinical use, objective biomechanical comparisons between DC-F and DC-PRF remain limited, and their relative *in vivo* stability and long-term performance under physiological mechanical loading are not well understood.

Therefore, this study aims to comparatively evaluate the biomechanical stability of DC-F and DC-PRF grafts using finite element analysis (FEA). The present study was guided by two research questions: 1. How do the stress distributions and deformation patterns of DC-F and DC-PRF grafts differ under daily and impact loading conditions during postoperative healing? 2. How does healing progression (immediate, 1 month, and 1 year) influence the mechanical stability of each graft type? Finite element models were developed in ANSYS to represent each graft type, and simulations were performed at multiple postoperative time periods to account for healing and tissue integration, with material properties updated based on literature-reported data. Comparative analysis of early and long-term stability provides a quantitative framework to support surgical decision-making, with the potential to improve aesthetic outcomes and reduce revision rates. While previous clinical studies have compared DC-F and DC-PRF in terms of aesthetic outcomes, complication rates, and patient satisfaction, quantitative biomechanical comparisons remain limited. Most existing evaluations rely on retrospective clinical observations rather than predictive mechanical modeling. The present study introduces a time-dependent finite element framework that simulates healing-related changes in material properties and interface conditions, enabling objective comparison of graft stability under physiologically relevant

loading. This modeling-based approach provides mechanistic insights that complement clinical findings and support evidence-based surgical planning.

Finite Element Modeling

A three-dimensional (3D) finite element model (FEM) was developed based on ultrasound images of transverse and sagittal sections of the nasal dorsum acquired one month after augmentation surgery.⁷ The model represents two reconstruction conditions with DC-F and DC-PRF. The model geometry was constructed using ANSYS software and consists of five components: nasal soft tissue envelope (STE), graft, nasal bone, lateral cartilage, and septal cartilage. The upper portion of the nose was modeled as nasal bone, while the lower portion was modeled as lateral cartilage, as shown in Figure 1; the outermost layer represents the STE, and the region between the STE and the underlying bone and cartilage corresponds to the DC graft. While the nasal structure is symmetric, only one half of the geometry was simulated to reduce computational cost. A symmetric boundary condition was applied along the plane of symmetry. The interfaces between the bone/cartilage were assumed bonded and the facial structure were assumed to be immobile. Fixed boundary conditions were therefore applied for the bottom of the bone and cartilage. Specifically for the DC-F model, fixation is achieved by anchoring the fascia strip at two edges on the nasal root's golden point and the lateral cartilage to ensure stable positioning.⁴ The interfaces between the DC graft and the bone or cartilage depend on the graft type and the postoperative time point, as described in detail in the following section. The material properties of the skin, bone, and cartilage used in the simulations are listed in Table 1. For simplicity, these materials were modeled as linear elastic. Assuming linear elastic allows for a focused comparative analysis of the geometry and material stiffness effects between the two graft types.

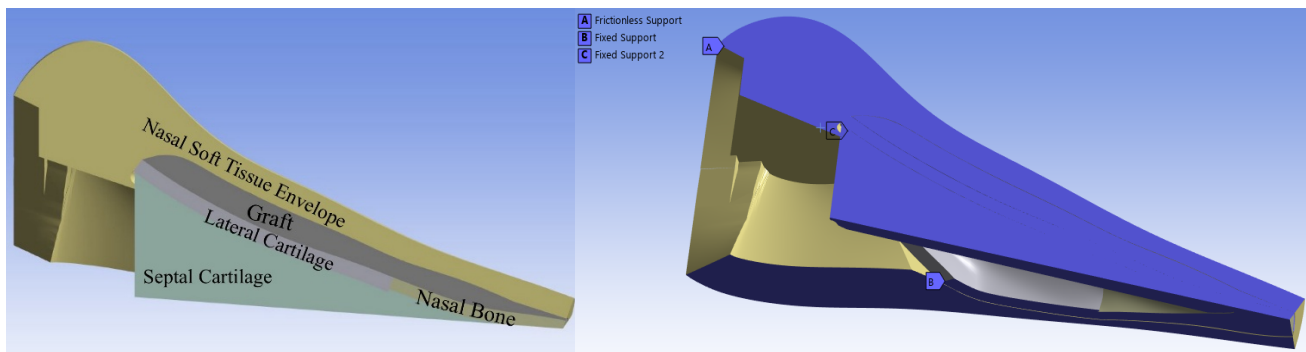


Figure 1: Geometry of the FEM (left) and boundary conditions (right)

Table 1: Mechanical properties of nasal bone, septal cartilage, lateral cartilage, costal cartilage and nasal soft tissue envelope.⁸⁻¹¹

	Young's Modulus (MPa)	Poisson's Ratio
Nasal Bone	15000	0.30
Septal Cartilage	2.9	0.30
Lateral Cartilage	0.98	0.30
Costal Cartilage	10	0.30
Nasal Soft Tissue Envelope	0.135	0.48

Following the surgical procedure, the healing process can extend beyond one year.⁷ This process involves tissue integration. The interfaces between the DC–F graft and the surrounding bone or cartilage vary with postoperative time because the outer layer of the graft consists of fascia lata. At the initial stage, there is minimal connection between the fascia lata and the adjacent nasal tissues. Over time, tissue integration begins, followed by initial adhesion and eventual bonding. This process was simulated by varying the coefficient of friction (COF) from 0.3 to 0.8 at the interface, with the final case fully bonded as shown in Table 2. In contrast, for the DC–PRF graft, the PRF matrix is a highly adhesive gel; therefore, the graft is assumed to be bonded to the surrounding tissues from the immediate postoperative stage.

For the mechanical properties of the grafts, Poisson's ratio was assumed to be 0.4, which lies within the typical range for biological soft tissues.¹² For DC–F, the diced costal cartilage particles are wrapped within a fascia lata envelope, resulting in a relatively stiff and solid structure. For DC–PRF, the graft is comparatively soft because the diced cartilage particles are embedded within a compliant PRF matrix. Because DC–PRF techniques vary by surgeon and application, there is no fixed ratio for mixing diced cartilage with the PRF matrix. In this study, two cartilage volume fractions, 40% and 80%, were assumed. The initial Young's modulus was calculated using the rule of mixtures for particle-reinforced composite materials, as shown in Equations 1-2. The Young's modulus of costal cartilage and PRF were taken as 4 MPa and 0.05 MPa, respectively, and the effective modulus was assumed to be the midpoint between the upper and lower limits.^{5, 13} As the graft matrix is gradually absorbed and replaced by fibrous connective tissue over time, the graft becomes progressively more solid and stiffer. As reported by Kulaksiz et al., 1.1 to 1.2 increase after one month of healing for DC-F and 2-3 times for DC-PRF after one month post-surgery.⁵ Both graft types have been reported to eventually reach a stable level comparable to that of solid cartilage.

Rule of Mixture for elastic modulus (E) of DC-PRF:

$$\text{Upper limit: } E_{upper} = E_m V_m + E_p V_p \quad (1)$$

$$\text{Lower Limit: } \frac{1}{E_{lower}} = \frac{V_m}{E_m} + \frac{V_p}{E_p} \quad (2)$$

where E represents Young's modulus, V denotes the volume fraction, with subscripts m and p referring to the PRF matrix and DC particles, respectively. The interface contact conditions, and Young's modulus values are summarized in Table 2 for the three postoperative stages considered: immediate postoperation (T1), one month postoperation (T2), and one year postoperation (T3).

Table 2: Interface contact conditions and Young's modulus values for DC-F and DC-PRF at three healing stages.^{4,5,7,14}

Time Period		Interface Contact	Young's Modulus (MPa)
T1	DC-F	Frictional: COF 0.3	3.5
	DC-PRF	Bonded	40% DC: 0.86 80% DC:1.72
T2	DC-F	Frictional: COF 0.8	3.85
	DC-PRF	Bonded	40% DC: 1.72 80% DC:3.44
T3	DC-F	Bonded	4
	DC-PRF	Bonded	4

For the loading condition, since the objective of the present study is to compare the mechanical responses of the two graft types, a load of 0.5 N was applied to the surface of the nasal model to represent daily constant external loading, such as that induced by eyeglasses. A nose-pad force of 0.5N was used to represent a conservative upper-bound loading case for eyeglass wear.¹⁵ Another scenario is that some object impacts the nose. A force, 5N, is applied on the bridge of the nose model as shown in Figure 2.

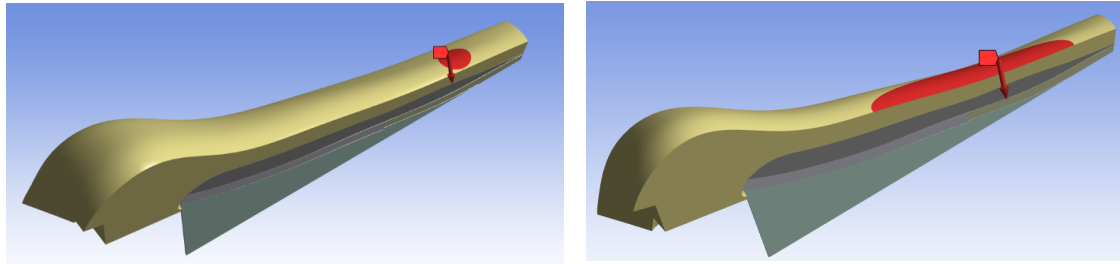


Figure 2: Two loading scenarios were considered for the model: daily constant external loading of 0.5 N (left) and an impact load of 5 N (right).

Results and Discussion

Numerical simulations were performed for three time periods for models incorporating each graft type. Mesh convergence was verified based on the maximum von Mises stress. The maximum von Mises stress results for the different graft configurations—DC-F, DC-PRF (40% DC), and DC-PRF (80% DC)—under two loading conditions, daily constant external loading and impact loading, are summarized in Table 3 and illustrated in Figure 3. The von Mises stress distributions in the grafts for DC-F and DC-PRF with 40% DC are shown in the Appendices. The results show that, for DC-PRF grafts, the DC-to-PRF ratio affects stress levels, with the largest difference observed at T2 under impact loading (21.3% increase). The DC-F graft exhibits substantially higher maximum von Mises stresses than all DC-PRF models. At T2 under daily constant external loading, stresses in the DC-F graft are approximately 38 and 36 times higher than those in DC-PRF grafts with 40% and 80% DC, respectively. Under impact loading, the largest differences also occur at T2, with DC-F stresses about 12 times higher than DC-PRF with 40% DC and 10 times higher than DC-PRF with 80% DC. Overall, the higher stress levels in DC-F indicate a greater failure risk. For DC-PRF grafts, stress increases from T1 to T3 as graft

stiffness increases over time. In contrast, DC-F reaches peak stress at T2 under both loading conditions prior to full bonding; once bonded at T3, stress levels decrease. This indicates that T2, approximately one month after surgery, represents a critical period during which stresses reach their maximum and the healing process should be carefully monitored. Clinically, elevated stress concentrations may increase the risk of graft displacement, contour irregularities, or resorption. The substantially higher stresses observed in DC-F at T2 suggest that patients may benefit from reduced mechanical loading (e.g., avoiding eyeglass wear or nasal pressure) during this critical healing phase.

Table 3: Maximum von Mises stress in DC-F, DC-PRF (40% DC) and DC-PRF (80% DC) under two loading conditions, daily constant external loading and impact loading

Time Period	Stress Under Daily Constant Load (MPa)			Stress Under Impact (MPa)		
	DC-F	DC-PRF 40%	DC-PRF 80%	DC-F	DC-PRF 40%	DC-PRF 80%
1	1.88	0.058	0.062	1.92	0.169	0.188
2	2.34	0.062	0.065	2.26	0.188	0.228
3	0.43	0.065	0.065	1.09	0.246	0.246

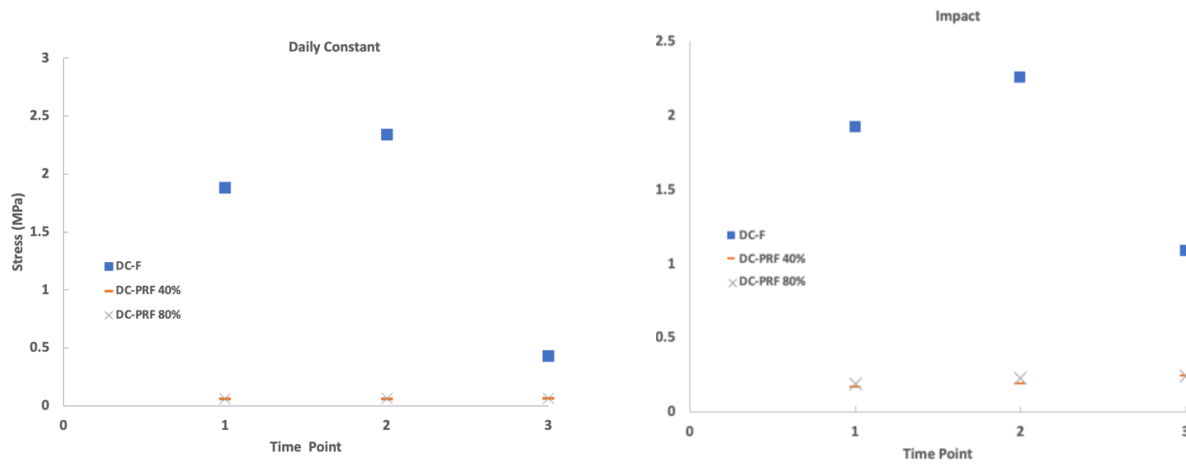


Figure 3: Maximum von Mises stress in DC-F, DC-PRF (40% DC) and DC-PRF (80% DC) under two loading conditions, daily constant external loading (left) and impact loading (right)

To evaluate graft responses under different loading conditions, the results were reorganized and plotted in Figure 4. For the DC-F graft, loading conditions have minimal effect at T1 and T2 because the graft has not yet bonded to the surrounding tissue. Once bonding occurs at T3, impact loading produces approximately 1.5 times higher stress. In contrast, for DC-PRF grafts, impact loading results in stresses approximately 3.0, 3.5, and 3.8 times higher than daily constant loading at T1, T2, and T3, respectively, indicating that DC-PRF grafts are more sensitive to loading type than DC-F.

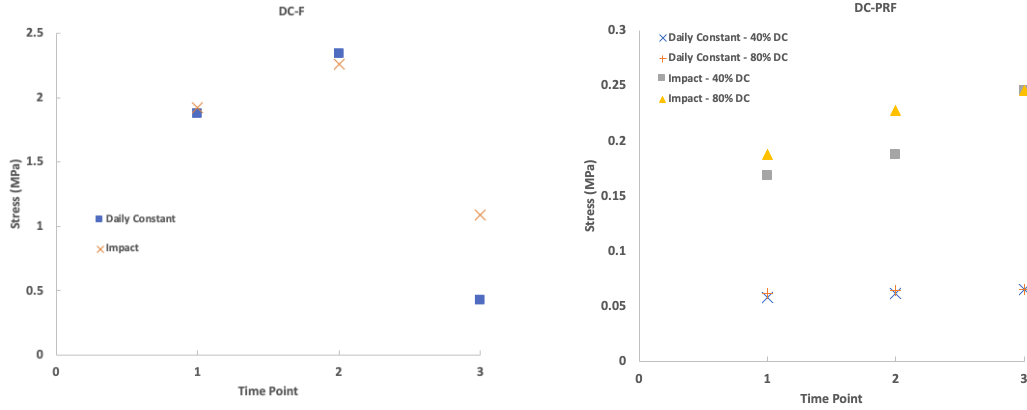


Figure 4: The maximum von Mises stress in the models under the two loading scenarios: daily constant external loading and Impact

Maximum deformations for the different graft types under the two loading conditions are summarized in Table 4. Under daily constant loading, all grafts show decreasing deformation over time, with DC-F consistently exhibiting the largest values, while DC-PRF grafts deform less, particularly at higher DC content. Under impact loading, deformations are generally higher, and DC-PRF grafts show greater sensitivity early in healing (T1 and T2), with DC-PRF 40% reaching the highest values. By T3, deformations of all grafts under both loading conditions converge, reflecting integration with the surrounding tissue. Increased deformation under impact loading may correspond to transient dorsal contour changes or graft micromotion, which could compromise aesthetic outcomes. The higher early deformation observed in DC-PRF grafts suggests a need for protective measures during early healing.

Table 4: Maximum deformations for the different graft types under the two loading conditions

Time Period	Deformation Under Daily Constant Load (mm)			Deformation Under Impact (mm)		
	DC-F	DC-PRF 40%	DC-PRF 80%	DC-F	DC-PRF 40%	DC-PRF 80%
1	0.159	0.074	0.040	0.201	0.393	0.278
2	0.142	0.040	0.021	0.194	0.278	0.215
3	0.043	0.018	0.018	0.224	0.206	0.206

Conclusions

The study provides quantitative insights into the mechanical stability and behavior of DC-F and DC-PRF grafts, supporting surgical decision-making. DC-F offers greater initial structural stability but may pose a higher risk under peak stresses, while DC-PRF provides a more compliant, impact-sensitive alternative. Understanding these differences can help optimize graft selection, placement, and postoperative monitoring to improve aesthetic outcomes and reduce revision rates in rhinoplasty. Beyond its clinical relevance, this study highlights finite element modeling as a valuable tool for biomedical engineering practice and education. The integration of patient-specific geometry, time-dependent material properties, and healing-phase interfaces illustrates how computational modeling supports interdisciplinary collaboration and can serve as an educational case study connecting mechanics, biomaterials, and clinical decision-making.

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Appendices

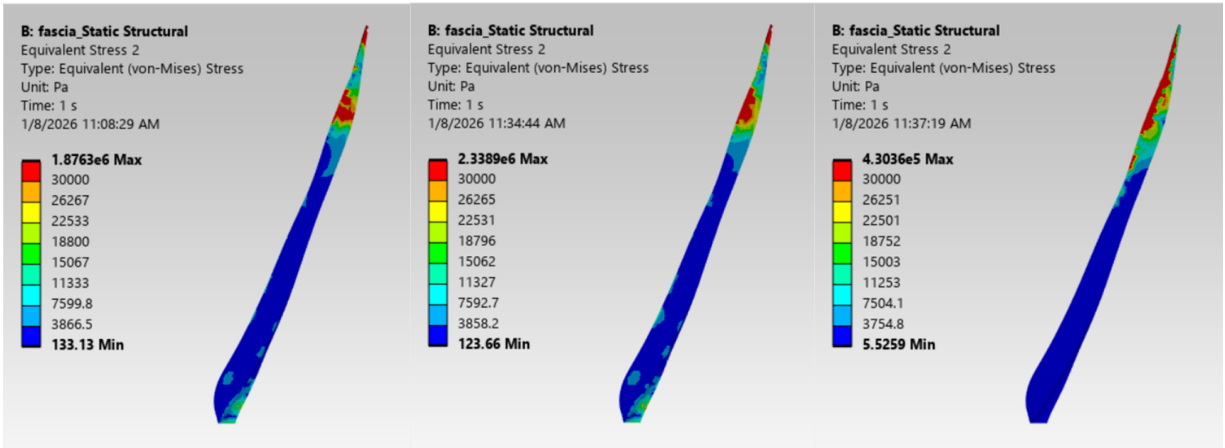


Figure 5: Von Mises stress distributions in the DC-F graft at T1–T3 under daily constant load

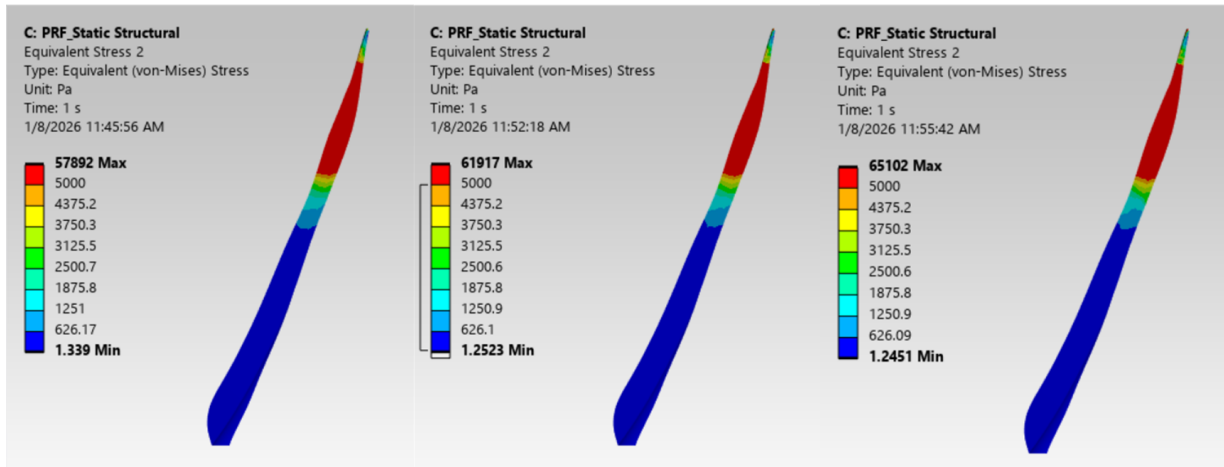


Figure 6: Von Mises stress distributions in the DC-PRF graft with 40% DC at T1–T3 under daily constant load

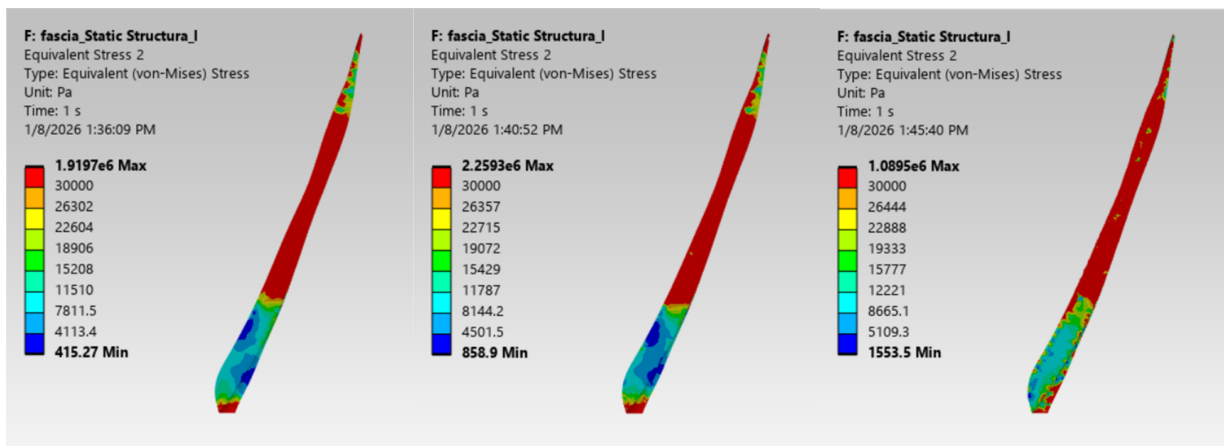


Figure 7: Von Mises stress distributions in the DC-F graft at T1–T3 under impact load

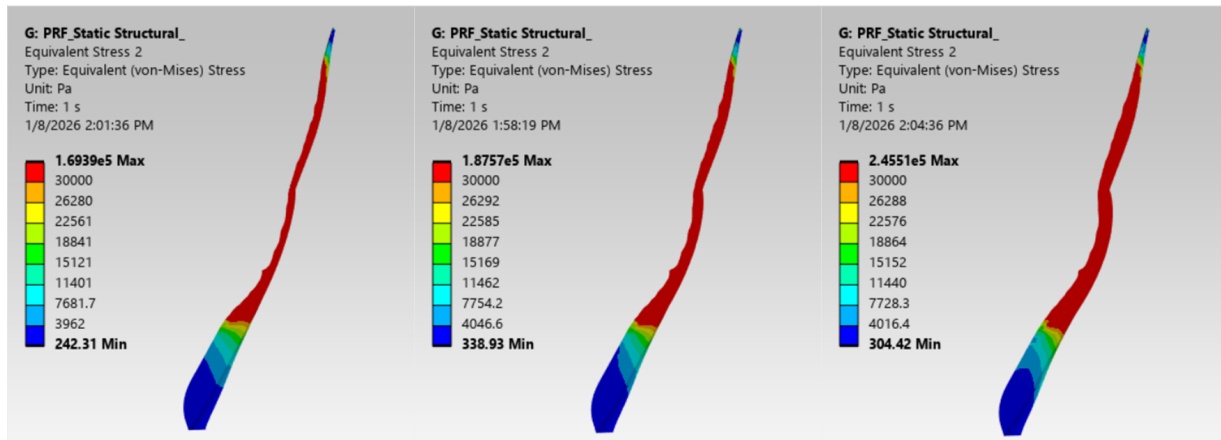


Figure 8: Von Mises stress distributions in the DC-PRF graft with 40% DC at T1–T3 under impact load